

Complete Guide to package.json in Node.js & Playwright

package.json is the main configuration file for Node.js projects. It manages project metadata, dependencies, scripts, versions, and configurations.

1. What is package.json?

package.json is a configuration file used in Node.js applications. It acts as the central control file for managing project information and dependencies.

2. Why package.json is Important

- Manages dependencies
- Defines scripts
- Stores project metadata
- Enables version control
- Makes projects shareable and reproducible

3. Creating package.json

Use the following commands:

4. Basic Structure

A typical package.json contains metadata, scripts, dependencies, and configurations.

5. Main Fields

Important fields include name, version, description, scripts, dependencies, and devDependencies.

6. Dependencies vs DevDependencies

Dependencies are required at runtime. DevDependencies are only needed during development/testing.

7. npm Scripts

Scripts help automate tasks such as testing, building, linting, and running applications.

8. Playwright Example

Playwright projects use package.json to manage browser automation dependencies and test scripts.

9. package-lock.json

package-lock.json stores exact dependency versions for consistent installations.

10. Best Practices

- Use meaningful scripts
- Keep dependencies updated
- Separate devDependencies
- Use semantic versioning
- Avoid unnecessary packages

11. Common npm Commands

Useful npm commands for package management.

12. Interview Questions

Frequently asked interview questions related to package.json.

13. Important Commands

Initialize package.json

```
npm init
```

Quick Initialization

```
npm init -y
```

Install Dependency

```
npm install lodash
```

Install Dev Dependency

```
npm install -D typescript
```

Run Script

```
npm run test
```

14. Sample package.json

```
{  
  "name": "playwright-framework",  
  "version": "1.0.0",  
  "description": "Playwright automation framework",  
}
```

```

    "scripts": {
      "test": "playwright test",
      "headed": "playwright test --headed"
    },
    "dependencies": {
      "pdf-parse": "^1.1.1"
    },
    "devDependencies": {
      "@playwright/test": "^1.52.0",
      "typescript": "^5.8.0"
    }
  }
}

```

15. package.json vs package-lock.json

File	Purpose
package.json	Stores dependency ranges and project metadata
package-lock.json	Stores exact installed dependency versions

16. Interview Questions

- What is package.json?
- Difference between dependencies and devDependencies?
- What is package-lock.json?
- How do you create package.json?
- How do you run npm scripts?

Summary:

package.json is one of the most important files in Node.js and Playwright projects. It manages dependencies, scripts, metadata, and project configurations, enabling efficient development workflows.